



CampusRide

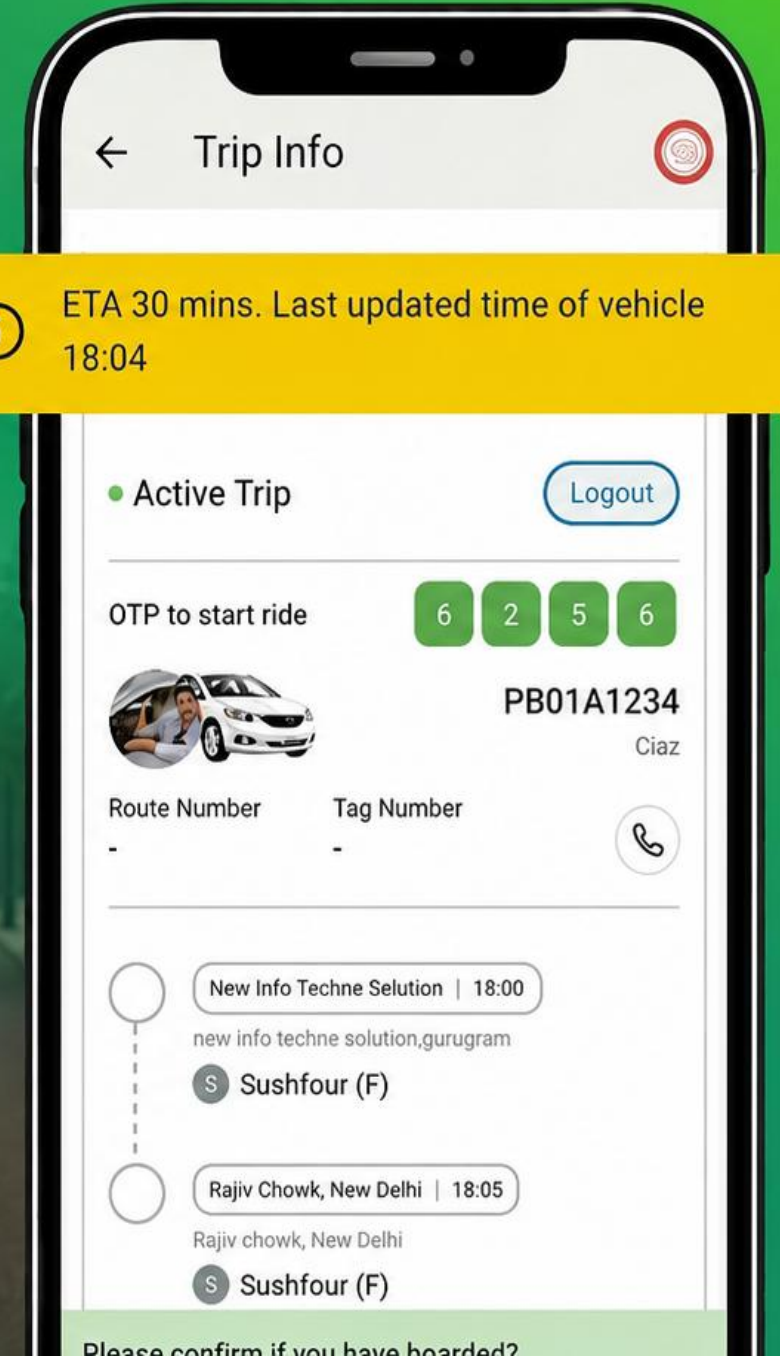
Real-Time Campus Mobility and Ride Management Platform

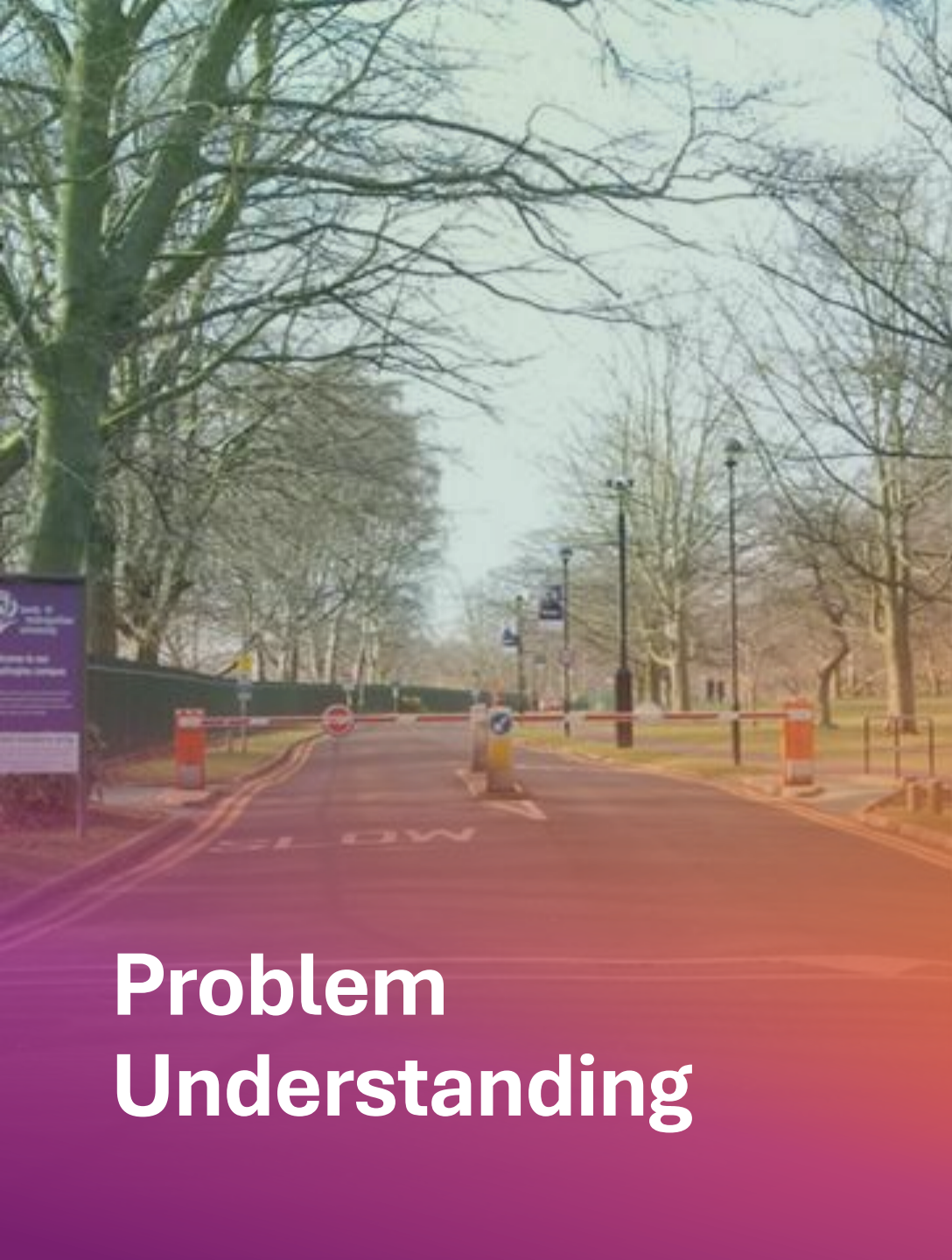
Submitted by

Abhinav Kumar

Kalpesh Nandu Jagtap

Aditya Kumar Gaurav





Problem Understanding

- **Problem Statement**

- Large campuses depend heavily on e-rickshaws for transportation.
- Current ride coordination is: Informal and fragmented

- Time-consuming

- Inefficient for passengers

- Difficult for drivers to manage demand

- **Proposed Solution**

- Our platform provides: Passenger registration and ride requests

- Driver onboarding and availability management

- Live ride tracking

- Ratings and feedback

- Analytics dashboard

- **Key Objectives**

- Improve transportation efficiency
- Reduce passenger waiting time
- Increase driver utilization
- Enable real-time communication

System Architecture

Passenger App (React)



REST APIs



Node.js + Express Server



Socket.IO



MongoDB



Driver App (React)

- **Components**
- Frontend: React, React Router, Recharts, Leaflet Maps
- Backend: Express.js, Socket.IO, JWT Authentication
- Database: MongoDB + Mongoose

Database Schema

User Collection

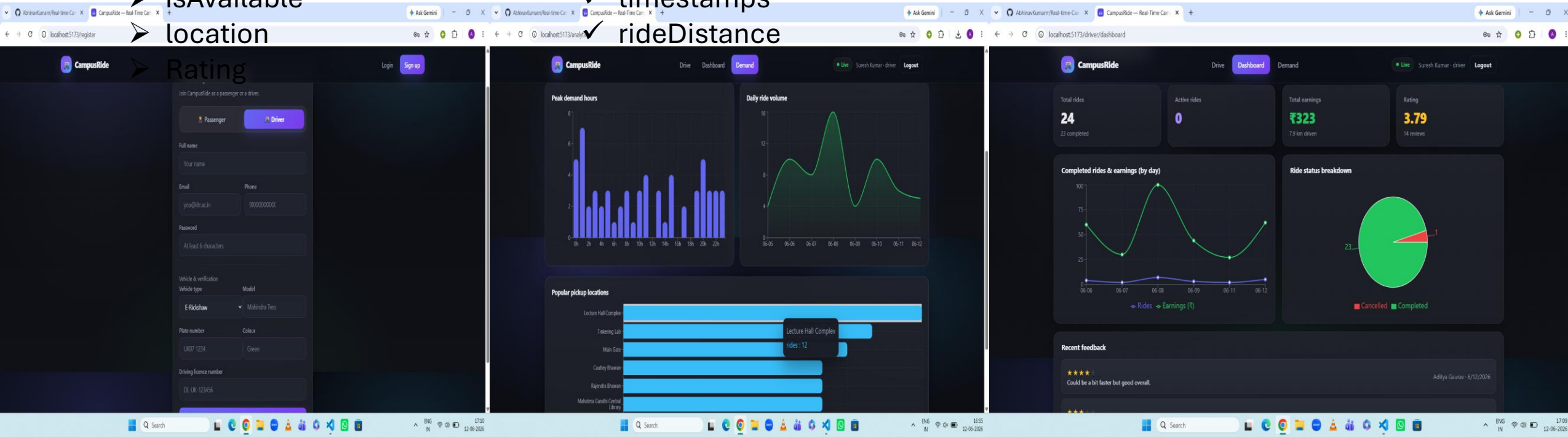
- name
- email
- password
- role
- vehicleInfo
- isAvailable
- location

Ride Collection

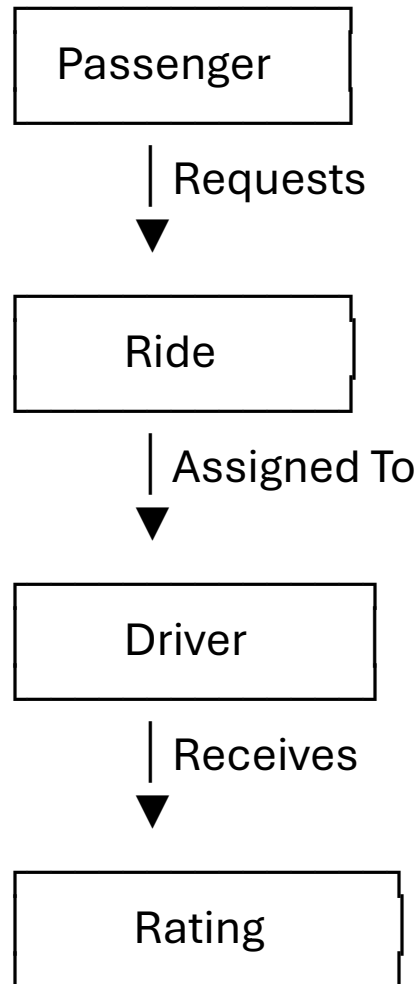
- ✓ passengerId
- ✓ driverId
- ✓ pickup
- ✓ destination
- ✓ status
- ✓ timestamps
- ✓ rideDistance

Rating Collection

- ❖ rideId
- ❖ passengerId
- ❖ driverId
- ❖ rating
- ❖ feedback



Entity Relationship Diagram (ERD)



Relationships

- One Passenger → Many Rides
- One Driver → Many Rides
- One Ride → One Rating
- One Passenger → Many Ratings
- One Driver → Receives Many Ratings

API Overview

Authentication APIs

- POST /auth/register
- POST /auth/login
- GET /auth/me

Driver APIs

- GET /drivers/available
- PATCH /drivers/availability
- PATCH /drivers/location

Ride APIs

- POST /rides
- GET /rides
- PATCH /rides/:id/accept
- PATCH /rides/:id/start
- PATCH /rides/:id/complete
- PATCH /rides/:id/cancel

Rating APIs

- POST /ratings
- GET /ratings/driver/:id

Analytics APIs

- GET /analytics/driver
- GET /analytics/demand



Driver-Console communication

Accelerate actions and manage your active ride.



Socket.IO

Real-time, bi-directional communication



Instant ride updates

Get notified instantly on any ride activity



Driver availability synchronization

Live sync driver status across the platform



Live ride lifecycle tracking

Track ride progress from start to completion



Authentication

Secure login for both drivers and passengers



JWT

Token-based authentication for secure APIs



Stateless

Scalable and server-independent sessions



Secure

Data encryption and secure communication



Scalable

Built to handle growing users and requests

Active ride

● In Progress



Passenger Picked



Ride in Progress

4

Completed

Passenger

Abhinav Kumar

Route

Mahatma Gandhi Central Library

Gate

Gate 1

Vehicle

MH12 AB 1234

Complete ride

Cancel

Incoming requests

0 0



Waiting for requests...

Status

Online • Available

Go Offline



Database

Optimized location



MongoDB

High performance & flexible



Flexible schema

Easily adaptable to changes



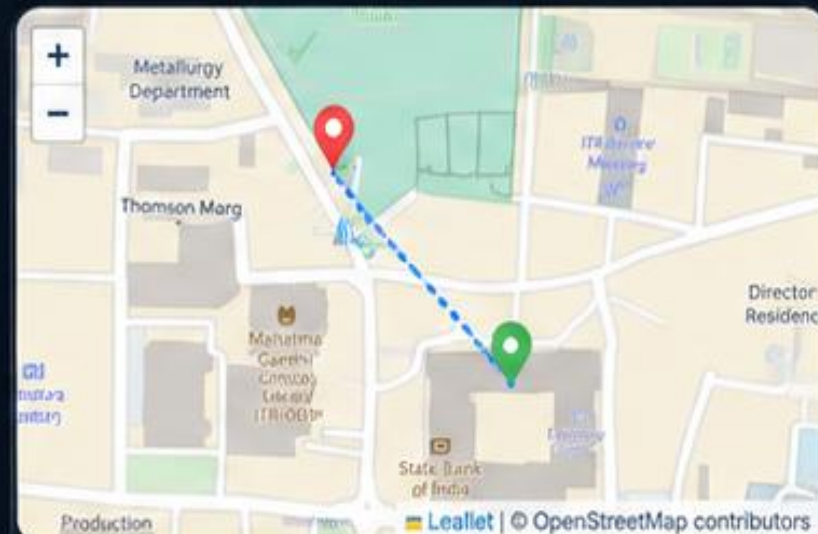
Fast development

Rapid feature implementation



Easy Node.js integration

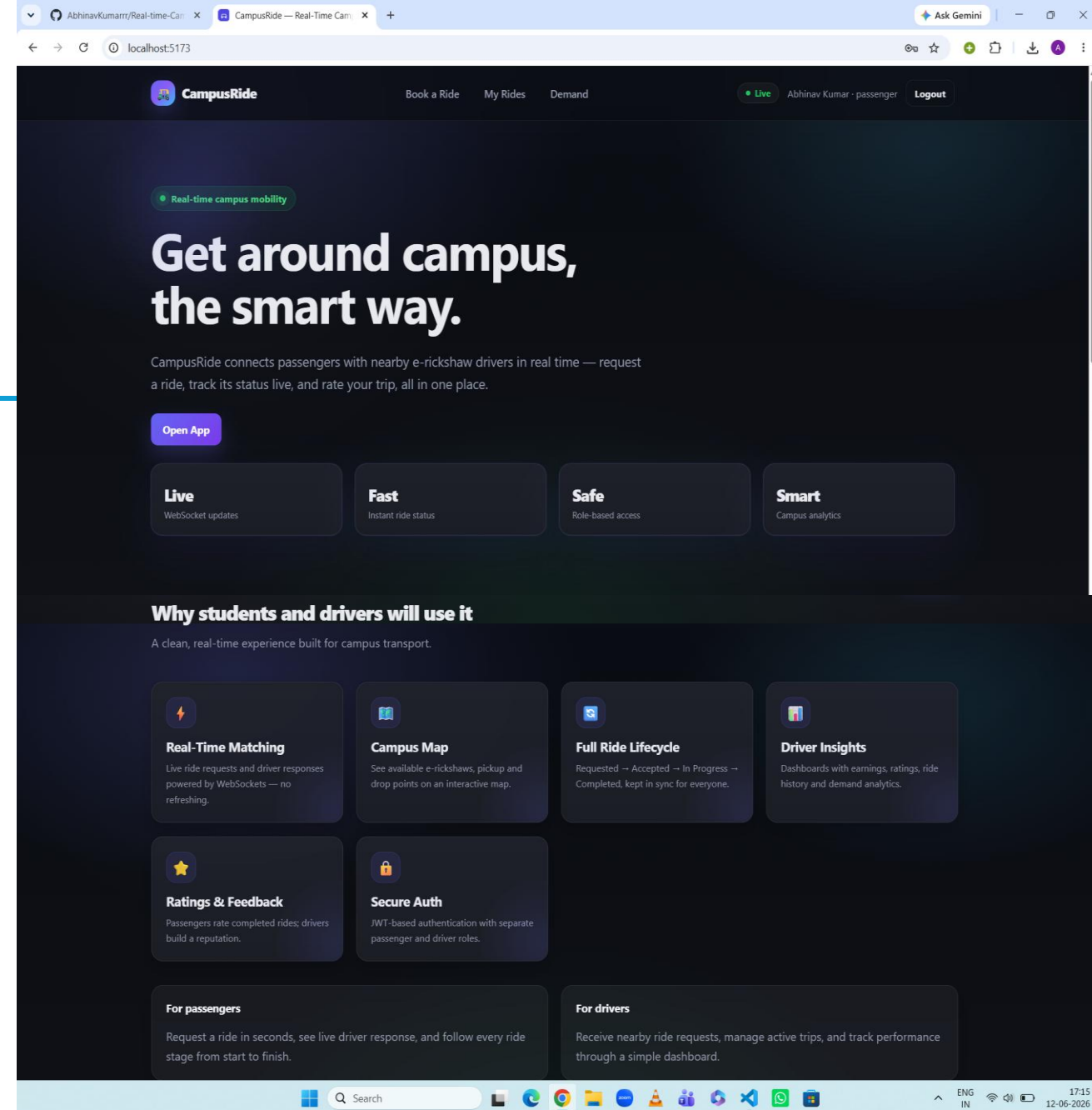
Smooth backend connectivity



● You ● Pickup ● Drop

Features

- ✓ Authentication & Profile Management
- ✓ Driver Availability Management
- ✓ Ride Request Workflow
- ✓ Real-Time Ride Updates
- ✓ Ride Lifecycle Management
- ✓ Driver Dashboard
- ✓ Ratings & Feedback
- ✓ Live Campus Map (Leaflet + OpenStreetMap)
- ✓ Demand Analytics Dashboard





Thank You!



Thank you for being a part of the CampusRide journey.

Your trust and support drive us to build smarter, safer and more connected campus mobility experiences every day.



Safe & Secure

Ensuring every ride is safe and reliable.



Real-Time Updates

Stay informed with live tracking and updates.



Smart Integration

Seamless systems for better coordination.



Sustainable Future

Building eco-friendly and efficient mobility.



Better Experience

Continuously improving for your convenience.



Need Help?

Our support team is always here for you.



+91 98765 43210



support@campusride.com

Go to Dashboard →

Let's continue building a smarter and better campus together! 🚀



Made with ❤️ for a better tomorrow

